0. **Code:** Janumala Suraaj- 2254078.zip

1. Final Test / Final Test B / Program.cs
2. Final Test / Final Test C / Program.cs
3. Final Test / Final Test / Program.cs

1. Final Test / Final Test B / Program.cs

```
namespace Final_Test_B
{
    internal class Program
    {
        static void Main(string[] args)
        {
            //Create an integer array
            int[] array = {
                1,
                10,
                23,
                5,
                2
            };
            int[] countArray = new int[array.Length];
            //call the function for insertion sort
            InsertionSort(array, countArray);
            //Iterate through the sorted array and print each number
            Console.WriteLine("Sorted Array: ");
            for (int i = 0; i < array.Length; i++)
            {
                Console.WriteLine(array[i] + " ");
            }
            Console.WriteLine();

            Console.WriteLine("count Array: ");
            for (int i = 0; i < countArray.Length; i++)
            {
                Console.WriteLine(countArray[i] + " ");
            }
            Console.WriteLine();
```

Create this in your function

B2

B1

1. **Final Test / Final Test B / Program.cs**

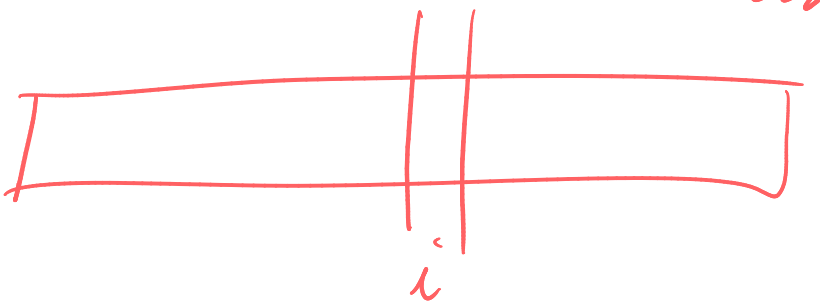
```
        }
        //Create a function for insertion sort
        public static void InsertionSort(int[] array, int[]
          countArray)
        {
            //iterate through the array starting from second element
            for (int i = 1; i < array.Length; i++)
            {
                int key = array[i];
                int j = i - 1;
                int count = 0;
                //compare each element with element before it and if current
                  element is smaller than the element before it shift larger
                  element to right and current element to its position
                while (j ≥ 0 && array[j] > key)
                {
                    array[j + 1] = array[j];
                    j--;
                    count++;
                }
                array[j + 1] = key;
                countArray[i] = count;
            }
        }

    }
}
```

B1

B3

Insertion Sort Implementation

B4 Sorting an array



2. Final Test / Final Test C / Program.cs

```
using System.Diagnostics.CodeAnalysis;

namespace Final_Test_C
{
    internal class Program
    {
        static void Main(string[] args)
        {
            //Create an array
            int[] array = {
                1,
                2,
                3,
                2,
                7,
                3,
                4,
                4,
                5,
                5,
                4,
                5,
                1,
                6,
                1,
                1
            };
            //call the function for finding sum of largest repeated
              numbers
            int result = SumofLargestRepeatedValues(array);
```

2. Final Test / Final Test C / Program.cs

```
30          Console.WriteLine($"Sum of all the largest numbers repeated
              in every different frequency is {result}");
31      }
32      // Create the function to find the sum of largest repeated
          numbers
33      static int SumofLargestRepeatedValues(int[] array)
34      {
35          //if input is null or empty return 0
36          if (array == null || array.Length == 0)
37          {
38              return 0;
39          }
40          //Create a dictionary for occurences
41          Dictionary < int, int > occurenceDict = new Dictionary < int,
              int > ();
42
43          //count the occurences
44          foreach(int i in array)
45          {
46              if (occurenceDict.ContainsKey(i))
47              {
48                  occurenceDict[i]++;
49              }
50              else
51              {
52                  occurenceDict[i] = 1;
53              }
54          }
55          //create a max dictionary
56          Dictionary < int, int > maxDict = new Dictionary < int, int >
              ();
```

C1

C5

num → count

C2

2. Final Test / Final Test C / Program.cs

```
            //find the largest number for each frequency
            foreach(var rep in occurenceDict)
            {
                int number = rep.Key;
                int frequency = rep.Value;
                if (!maxDict.ContainsKey(frequency) || number >
                  maxDict[frequency])
                {
                    maxDict[frequency] = number;
                }
            }
            int sum = 0;
            foreach(var rep in maxDict)
            {
                sum += rep.Key * rep.Value;
            }
            return sum;
        }


    }
}
```

count > max num

C3

C4

C6 good.

3. Final Test / Final Test / Program.cs

```
namespace Final_Test
{
    internal class Program
    {
        static void Main(string[] args)
        {
            //Declare integer and basenumber
            int number;
            int baseN;
            int sum = 0;
            int largestDigit = 0;
            //prompt user for integer and basenumber
            Console.Write("Enter the integer");
            number = Convert.ToInt32(Console.ReadLine());
            //ensure basenumber is positive using do while loop and
              prompting for input until positive
            do {
                Console.Write("Enter the Base-N value (must be positive)");
                baseN = Convert.ToInt32(Console.ReadLine());
            } while (baseN ≤ 0);
            //Dropping the sign in negative numbers
            if (number < 0)
            {
                number = -(number);
            }
            //Decode the number into base number digits
            while (number > 0)
            {
                int digit = number % baseN;
                sum += digit;
                //Finding the largest digit
```

A5

A1

A2

A3

A3

3. **Final Test / Final Test / Program.cs**

```
                if (digit > largestDigit)
                {
                    largestDigit = digit;
                }
                number /= baseN;
            }
            //display results
            Console.WriteLine($"Sum of base - {baseN} digits = {sum}.");
            Console.WriteLine($"Max base - {baseN} digit =
              {largestDigit}");
        }
    }
}
```

A5

A3